IP Addressing: Classes and Private IP Ranges

Simple English notes with examples, step-by-step calculations and Linux commands

1. What is an IP Address?

An **IP address** is the **address of a device** on a network, like a house address for a computer. It lets devices find each other and send data. This guide covers **IPv4**, the common type, like `192.168.1.10`.

Term	Simple meaning
IP address	The number that identifies a device on a network.
Octet	One of the four numbers in an IPv4 address (each is 0 to 255, 8 bits).
Network part	First part of the address: says **which network** (like the street).
Host part	Last part: says **which device** in that network (like the house number).
Subnet mask	Shows where the network part ends. Example: 255.255.255.0 (or /24).
Gateway / router	The device that connects your network to other networks and the internet.
Public IP	Unique on the internet. Given by your internet provider.
Private IP	Used only inside a home or office network. Not routable on the internet.
NAT	The router's trick of sharing one public IP among many private devices.

2. IPv4 Format and Binary

An IPv4 address has **32 bits**, written as **4 octets** separated by dots. Computers read it in binary (0 and 1). Each octet is 8 bits, so each number can be 0 to 255.

```
Decimal :   192   .   168   .    1    .   10
Binary  : 11000000 . 10101000 . 00000001 . 00001010
```

Converting a number to binary (step by step)

Write the bit values **128 64 32 16 8 4 2 1**. Put 1 under a value if you use it, 0 if not, until the total equals the number.

Number	128	64	32	16	8	4	2	1	Binary
192	1	1	0	0	0	0	0	0	11000000
168	1	0	1	0	1	0	0	0	10101000
10	0	0	0	0	1	0	1	0	00001010

Check: 192 = 128 + 64. 168 = 128 + 32 + 8. 10 = 8 + 2.

3. IP Address Classes

In the classful system, the **first octet** tells the class. The class decides how many bits are for the network and how many for hosts.

Class	First octet	Starts with bits	Default mask	Networks	Hosts per network	Use
A	1 - 126	0	255.0.0.0 (/8)	126	16,777,214	Very large networks
B	128 - 191	10	255.255.0.0 (/16)	16,384	65,534	Medium networks
C	192 - 223	110	255.255.255.0 (/24)	2,097,152	254	Small networks
D	224 - 239	1110	none	-	-	Multicast
E	240 - 255	1111	none	-	-	Experimental

Remember: 127.x.x.x is reserved for **loopback** (your own computer), so Class A stops at 126. Today networks use CIDR (flexible masks like /26), but classes are still taught and still used to explain private ranges.

Memory trick and host formula

- **A**: 1 to 126 **B**: 128 to 191 **C**: 192 to 223. Look only at the **first number**.
- Usable hosts = $\mathbf{2^h - 2}$, where h = number of host bits. Two are lost: the **network address** (all host bits 0) and the **broadcast address** (all host bits 1).
- Class C: 8 host bits, so $2^8 - 2 = \mathbf{254}$ hosts. Class B: 16 bits, so 65,534. Class A: 24 bits, so 16,777,214.

Classify these addresses

Address	First octet	Class	Public or private?
8.8.8.8	8	A	Public
10.1.2.3	10	A	Private
172.20.1.1	172	B	Private (172.16 - 172.31)
172.40.1.1	172	B	Public (outside 172.16 - 172.31)
192.168.10.5	192	C	Private
200.10.5.7	200	C	Public
224.0.0.5	224	D	Multicast
127.0.0.1	127	(loopback)	Your own computer

4. Private IP Address Ranges

Private ranges are reserved for **inside** networks: homes, offices, schools and data centers. Routers on the internet do not forward them, so many networks can reuse the same private addresses.

Range	Class	From	To	Mask	Addresses
Class A private	A	10.0.0.0	10.255.255.255	/8	16,777,216
Class B private	B	172.16.0.0	172.31.255.255	/12	1,048,576
Class C private	C	192.168.0.0	192.168.255.255	/16	65,536

From your video: Class C private range: 192.168.0.0 to 192.168.255.255. It contains 256 separate /24 networks, such as 192.168.0.x, 192.168.1.x ... 192.168.255.x, each with 254 hosts. Most home routers use **192.168.0.x** or **192.168.1.x**.

Why private addresses? How NAT works

There are not enough public IPv4 addresses for every device. So devices at home use private IPs, and the router (with **one public IP**) translates them. This is **NAT**.

```
Phone     192.168.1.20 \
Laptop    192.168.1.21  --> Router (private 192.168.1.1) --> Internet
TV        192.168.1.22 /          public IP: 49.36.10.5

All three devices appear on the internet as 49.36.10.5
```

Fun fact: A neighbour can have the same 192.168.1.20. No clash occurs, because private addresses are never seen on the internet.

Other special addresses

Address	Meaning
127.0.0.1 (localhost)	Loopback: your own computer. Used to test networking.
0.0.0.0	'This host' or 'any address'. Used by servers listening on all interfaces.
169.254.x.x	Link-local (APIPA): a device gives itself this when DHCP fails. Means 'no proper IP received'.
x.x.x.255 (in a /24)	Broadcast: sends to all devices in that network.
255.255.255.255	Broadcast to the local network.

5. Worked Examples (Step by Step)

Example 1: 192.168.1.10 with mask 255.255.255.0 (/24)

Step	Result
1. Class (first octet 192)	Class C
2. Public or private?	Private (inside 192.168.0.0 - 192.168.255.255)
3. Network part / host part	192.168.1 / 10 (mask /24 = first 3 octets are the network)
4. Network address (host bits all 0)	192.168.1.0
5. Broadcast address (host bits all 1)	192.168.1.255
6. Usable host range	192.168.1.1 to 192.168.1.254 (254 hosts)

Example 2: 172.16.5.20 with mask 255.255.0.0 (/16)

Step	Result
Class	B (first octet 172)
Private?	Yes, 172.16 to 172.31 is private
Network address	172.16.0.0
Broadcast address	172.16.255.255
Usable hosts	172.16.0.1 to 172.16.255.254 (65,534 hosts)

Example 3: 10.1.2.3 with mask 255.0.0.0 (/8)

Step	Result
Class	A (first octet 10)
Private?	Yes, all of 10.x.x.x
Network address	10.0.0.0
Broadcast address	10.255.255.255
Usable hosts	10.0.0.1 to 10.255.255.254 (16,777,214 hosts)

Same network?: Two devices can talk directly only if their network part is the same. With /24: 192.168.1.10 and 192.168.1.50 are in the same network. 192.168.1.10 and 192.168.2.10 are **not**, so they need a router.

6. See IP Addresses on Linux

Syntax	Meaning
`ip -4 addr show`	Show your IPv4 addresses (the inet lines).
`hostname -I`	Print your IP addresses only.
`ip route`	Show routes, including the default gateway.

Syntax	Meaning
`ping -c COUNT ADDRESS`	Test if a device answers. -c = how many times.
`curl -s ifconfig.me`	Show your public IP (needs internet).

COPY AND PASTE: Check your addresses

```
ip -4 addr show
hostname -I
ip route
ping -c 3 127.0.0.1
curl -s ifconfig.me; echo
```

Command	What it does (in simple words)
`ip -4 addr show`	Lists network cards and their IPv4 address and mask.
`hostname -I`	Just the addresses: quick check of private IP.
`ip route`	Shows the gateway (router) your computer uses for the internet.
`ping -c 3 127.0.0.1`	Sends 3 test packets to yourself: proves the network software works.
`curl -s ifconfig.me; echo`	Asks a website for your public IP. Compare it with your private IP.

Output (what students will see)

```
2: eth0: <BROADCAST,MULTICAST,UP> ...
    inet 192.168.1.10/24 brd 192.168.1.255 scope global eth0   <-- IP /24, broadcast
192.168.1.10                                <-- hostname -I (private, Class C)
default via 192.168.1.1 dev eth0            <-- gateway (router)
3 packets transmitted, 3 received, 0% packet loss  <-- ping works
49.36.10.5                                  <-- public IP (differs from private)
```

Teach this: Look at the number after **inet**. If it starts with **192.168**, **10** or **172.16 to 172.31**, your computer has a **private** address and is behind NAT. The **/24** after it is the mask (255.255.255.0).

7. Cheat Sheet and Practice

Fact	Value
Class A / B / C first octet	1-126 / 128-191 / 192-223
Class D / E	224-239 multicast / 240-255 experimental
Private ranges	10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
Class C private	192.168.0.0 to 192.168.255.255
Loopback	127.0.0.1

Fact	Value
Usable hosts	2 to the power of host bits, minus 2
Network / broadcast address	Host bits all 0 / all 1

Practice questions

- Which class is 150.20.5.9? Is it public or private?
- Write the Class C private range.
- How many usable hosts are in a /24 network?
- Is 172.32.1.1 private?
- What are the network and broadcast addresses of 192.168.5.77/24?
- What does an address starting with 169.254 mean?
- Why can two homes both use 192.168.1.5?

Answers

No.	Answer
1	Class B (128-191), public.
2	192.168.0.0 to 192.168.255.255.
3	254.
4	No. Private Class B is only 172.16 to 172.31, so 172.32.x.x is public.
5	Network 192.168.5.0, broadcast 192.168.5.255.
6	The device failed to get an address from DHCP and gave itself a link-local address.
7	Private addresses are not used on the internet, so each home's NAT router keeps them separate.